

1. RESULTS

The primary experiment is the Class A search for a drifting, unresolved spectral carrier: 402 fine-channel windows toward 60 systems. The Class B search for an unresolved spectral excess is secondary and carries no drift discrimination: 1249 coarse windows. Together they are the 1651 windows of the released catalogue. **No carrier was found toward any of the 82 systems searched.** What follows is first how deep the search reaches, and then what it found and how each finding was disposed of.

1.1. Sensitivity: the power at which a carrier would have been found

The sensitivity of this survey is a measured completeness and not a threshold. Artificial carriers were added to the real calibrated visibilities, each at a random sub-channel phase and with a random drift rate from its window’s own grid, and recovered through the unchanged pipeline: extraction, baseline removal, de-drifted stack, trigger and the 512-position spatial screen. The power at which nine carriers in ten come back is EIRP₉₀, the only sensitivity quoted here. It charges for the whole chain, because a carrier counts as recovered only when it reaches the trigger and outranks every control.

Measured that way the completeness is not one number, and what it depends on is the control ring. In the 390 of 402 Class A windows whose ring is noise, nine carriers in ten are recovered at $3.45 P_{\text{trig}}$, where P_{trig} is that window’s own 5σ trigger power. In the other 12, toward β Pictoris, ALMA J1537–3319, HD 48370 and HD 139084, the ring carries resolved disc emission; a carrier there has to outshine the star’s own surroundings rather than the noise, and the injected ladder never reaches ninety per cent below $8 P_{\text{trig}}$, although every carrier is recovered by $20 P_{\text{trig}}$. The second figure is therefore a bound and not a value, and each window is quoted on the stratum it belongs to. A single factor fitted to both strata together would be 4.42, and that is the completeness of no window in the survey: it sits above the first stratum’s own spread and far below the second’s bound. Class B keeps one factor, $4.70 P_{\text{trig}}$: no injected coarse window has a bright ring, so the distinction cannot be measured there and the 4 coarse windows that do are flagged in the catalogue instead.

Separating the two strata moves power between windows; it does not make the survey better. The median Class A window reaches 3.35×10^{15} W of equivalent isotropic radiated power and the median system, on its best window, 1.77×10^{15} W, with a spread across systems of 7.74×10^{13} – 5.79×10^{16} W (Fig. 2a). The survey reaches the 12 m benchmark transmitter of 5.9×10^{14} W for 14 of the 60 systems and the round 10^{15} W level for 22. Against a single blanket factor those limits are about a fifth deeper for 390 windows and several times shallower for 12, and the worst-placed system in the survey moves the wrong way because its best window is one of them. A second term runs against us throughout: the assumed

TABLE 1

WHAT SETS EIRP₉₀, AND IN WHICH DIRECTION. THE TWO-SIDED TERMS COMBINE IN QUADRATURE TO ONE INTERVAL APPLYING TO EVERY LIMIT; THE SIGNED TERMS ARE BIASES, STATED WITH THEIR DIRECTION RATHER THAN FOLDED INTO AN ERROR BAR. THE CHANNEL RESPONSE IS CARRIED BY THE INJECTIONS RATHER THAN APPLIED AFTERWARDS: AT THE NINETY PER CENT POINT THE RECOVERED PEAK CHANNEL HOLDS $1.93\times$ THE NOMINAL TRIGGER, SO THE RESPONSE IS ALREADY INSIDE THE MEASURED COMPLETENESS AND APPLYING IT AGAIN WOULD COUNT IT TWICE.

	value	what it does to the limit
Measured completeness, as a multiple of a window’s own trigger power		
Ring is noise (390 windows)	$\times 3.45$	measured; $\times 3.37$ – $\times 4.20$ across resamples
Ring carries emission (12)	$> \times 8$	a bound: everything is recovered by $\times 20$, nothing at the trigger
One factor for both	$\times 4.42$	the completeness of no window; not used
Two-sided uncertainties (per cent), combined		in quadrature
Window-to-window transfer	$-22/ + 8$	the spread across the 21 injected windows
Flux, distance and visibility scale	± 9	calibration
Combined	$-23/ + 12$	$\times 0.77$ – $\times 1.12$ on every limit
Signed terms. A bias is not an error bar and is not added to one.		
Channel response	$\times 2.3$	carried by the injections, so already inside the completeness above
its residual	$+8$	the injected profile falls short of the exact response: conservative
Phase centre, annual parallax	$+1.5$	applied window by window; every limit shallower
Decorrelation	$-12/ + 17$	declared from the literature, not measured here

phase centre omits the annual parallax, so a nearby star observed in a long-baseline configuration loses amplitude. The loss is 0.13 per cent in the median window and 51.5 per cent at worst, passing a tenth in 76 of the 1650 windows that carry the astrometric keys, and dividing each window by the fraction it retains makes every limit shallower — the median Class A limit by 1.5 per cent. One window, Wolf 28 in A002_X9f2ff8_X25b0, has no such keys and is left uncorrected. Table 1 collects these signed terms and the two-sided ones, which combine to $\times 0.77$ – $\times 1.12$ on every limit.

Carriers were injected directly into 21 Class A windows and the result transferred to the remaining 381, which is the dominant caveat on all of the above. Figure 1(a) shows the individual measurements rather than their summary: 18 of the 21 resolve a ninety per cent point, between 2.9 and $7.4 P_{\text{trig}}$, and the three that do not are the brightest-ring windows of all. There is no dependence on observing band or on channel width, the median values differing by 0.2 and 0.3. There is an apparent difference between the two arrays, 3.3 against 4.5, and it is not the array: it is carried by 4 windows whose target has bright emission near the ring, and it falls to 3.3 against 3.4 once the star’s own leak into the control positions is removed. That is why the strata are cut on the ring. Removing the leak is a diagnostic and not the screen we apply, since it would make the test for a source depend on a model of that source; and it buys little, moving the noise-dominated figure to 3.38 and the disc-affected windows not at all, because there the ring is set by the disc. Only the distance is worse for a single star than for the sample: of the 77 Gaia parallaxes the worst fractional error is 1.65 per cent, toward γ Tri.

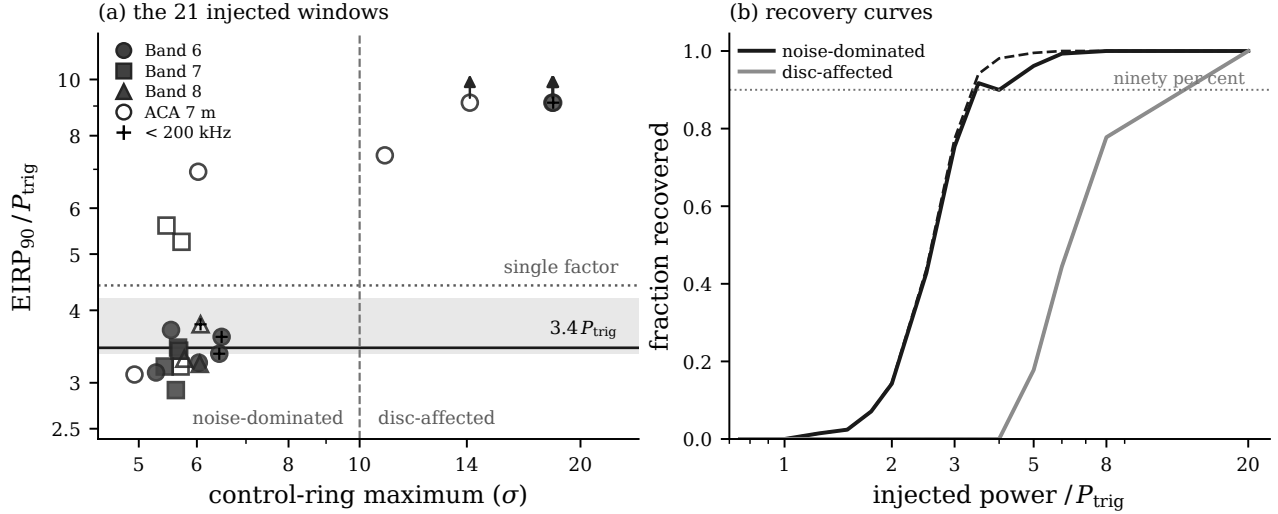


FIG. 1.— Why the completeness is quoted in two strata. (a) The 21 directly injected Class A windows, each at the power that recovers nine carriers in ten, against the maximum of its own control ring. Marker shape is the band, open symbols the compact array, a cross a channel narrower than 200 kHz, and arrows the three windows that never reach ninety per cent. The line and band are the noise-dominated value and its spread, the dotted line the single factor fitted to all of them. (b) The two strata’s recovery curves: solid as the screen is applied, dashed with the star’s own leak into the ring removed, which is a diagnostic and not the test used here. The disc-affected pair lie on top of one another.

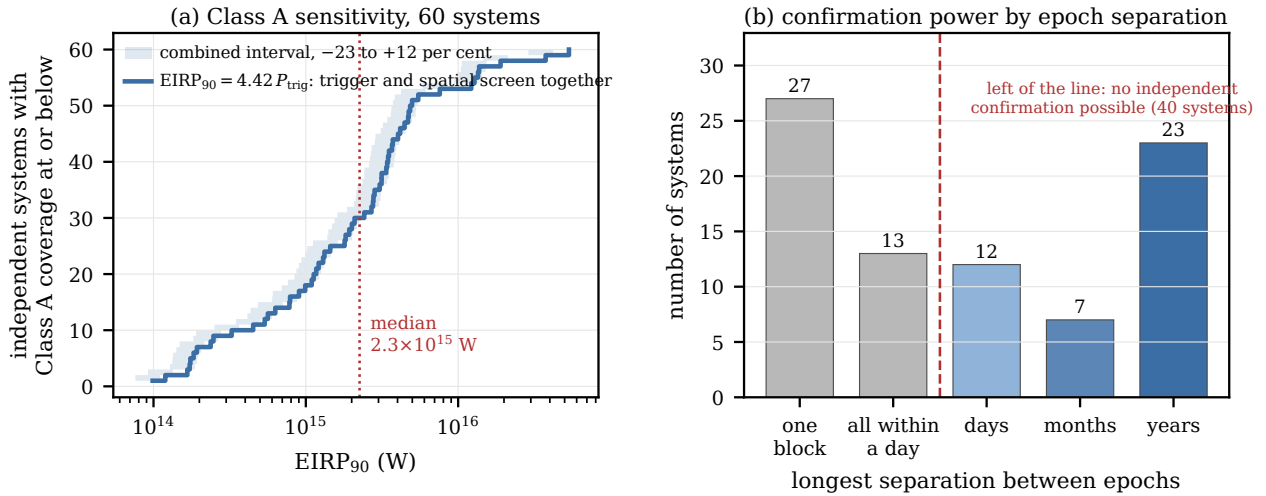


FIG. 2.— (a) Class A sensitivity: the cumulative number of systems whose best window reaches a given $EIRP_{90}$, the power at which nine carriers in ten are recovered through the trigger and the spatial screen together. The shaded band is the combined interval of Table 1. (b) The temporal selection function: systems grouped by the longest separation between their searched epochs, an independent epoch being a separation of more than 1 day. **For the 27 systems observed in a single block, and the 13 whose blocks all fall within one day — left of the dashed line — independent confirmation was not possible at all, whatever the sensitivity in panel (a).** Recurrence over years can be tested for 23 systems of 82.

1.2. The nearest stars: Barnard’s Star and Wolf 359

Barnard’s Star is the second-nearest stellar system, and Wolf 359 the third if brown dwarfs are excluded. To our knowledge these are the first millimetre or sub-millimetre technosignature limits for either; the searches checked are ?, ? and ?. Both are non-detections. *Both limits come from the secondary Class B experiment and not from the carrier search*: the archive holds only coarse Band 6 windows toward these two stars, so neither is searched for a drifting carrier and neither constrains one. What is constrained is an unresolved spectral excess. Quoted on the $EIRP_{90}$ convention of §1.1, Barnard’s Star reaches 3.0×10^{14} – 4.5×10^{14} W over 12 windows. Wolf 359 reaches 3.5×10^{14} – 5.1×10^{14} W over 12.

1.3. Threshold crossings and how each was disposed of

The result is a chain of four tests, and it is read in one direction. Of the 1651 windows, 56 contain a channel $\times$ drift cell at the stellar position reaching $T \geq 5$. Such a cell is a *threshold crossing*. The molecular-line mask, ± 50 km s $^{-1}$ wide, evaluated in each star’s own rest frame and fixed before any crossing was classified, identifies 19 of them with a catalogued transition. 37 are unattributed by that test, and every crossing is in one column or the other. Of the unattributed, 1 outranks all 512 control positions in its own window, and it does not recur in any later epoch covering its frequency. **No crossing survives the chain, and this survey reports no detection.** Figure 4 carries the chain with

TABLE 2

THE RESULT IN FIVE COUNTS, EACH READ FROM THE ROWS OF TABLE 5 SO THAT THE TWO CANNOT DISAGREE. THE LINE ATTRIBUTION IS EVALUATED IN THE STELLAR FRAME AT THE ADOPTED $\pm 50 \text{ km s}^{-1}$ HALF-WIDTH, AND THE LAST TWO LINES ARE THE SPATIAL SCREEN AND THE RECURRENCE TEST APPLIED IN THAT ORDER.

Quantity	Observed
Threshold crossings	56
Line-attributed ($ \Delta v \leq 50 \text{ km s}^{-1}$, stellar frame)	19
Unattributed	37
outranking all spatial controls	1
recurring at a second epoch	0

the number surviving each step, Table 2 the same five counts, and Table 5 the crossing-by-crossing ledger behind them.

The mask is informative rather than permissive, and its half-width is not a tuned parameter. The width follows from the Keplerian velocity bound on circumstellar gas in these systems, and Table ?? carries every count in the chain above at half-widths from $\pm$ to $\pm \text{km s}^{-1}$: the number outranking every control is the same at all of them, and it is the same crossing. The mask attributes 19 crossings where its occupancy of the searched band would give 2.3 by chance alone, so the attributions are a kinematic result and not an accident of bandwidth.

1.3.1. Most of the crossings are expected by chance

A 5σ trigger is a threshold on a statistic and not a false-alarm rate: the number of searched channel $\times$ drift cells per window spans several decades, so a common threshold buys very unequal protection. Summing each window's own measured false-alarm probability over the covered Class A windows gives 34.5 expected unattributed Class A crossings, against 35 observed. *About two thirds of the crossings this survey reports are expected by chance*, which is why the chain never treats a crossing as a detection and why no artefact hypothesis is needed to account for their number.

1.3.2. One execution block accounts for four of the crossings

4 crossings belong to BD+05 1668 (GJ 273, Luyten's Star), with the highest statistics in the survey outside β Pictoris and HD 48370. All 4 lie in one ACA execution block, A002_Xc7fa6f_X28a8, in which the whole field is above threshold: 512 of 512 control positions exceed $T = 5$, against a survey median control maximum of 2.75. The single-integration noise is as predicted (0.1609 Jy against 0.1611 Jy), but the residual does not integrate down; a jackknife on the channel mean gives 0.030 Jy against 0.0048 Jy thermal. A slow drift common to every channel and every position therefore inflates the statistic across the field. The 24 sibling executions of the same field, with the same array and the same spectral setup, reach a maximum of $T_\star = 4.26$ and produce no crossing. *The median of the control ensemble, not its maximum, is the quality criterion this kind of search needs.* We report it as a diagnostic and do not adopt it as a cut, because a cut strict enough to remove this block would also remove the HD 48370 carbon-monoxide control.

1.3.3. Stellar flares are excluded by a broadband prediction

The hosts are not a quiet sample: 2 of 2 carry activity indicators, and magnetically active stars do produce

coherent bursts at centimetre wavelengths. A millimetre flare, however, raises every channel of a window by the same flux density. A flare able to produce a 5σ single-channel crossing must therefore produce a $5\sqrt{N}\sigma$ continuum event in the same integrations. Over the 48 crossings with a recoverable continuum measurement that prediction runs $49\text{--}4,717\sigma$, while the observed continuum statistic has median -0.29 and $|Z| < 3$ in 45 of 48. One block settles it directly. AU Mic A002_Xa42f75_X319 contains both a crossing ($T_\star = 5.27$) and the published millimetre flare of ?. In the crossing's own time window the continuum is raised at $Z = 10.3$, while the crossing's de-drifted track reaches only $Z_{\text{max}} = 2.1$. *A real, bright, independently published millimetre flare in the same data contributes 0.17σ to the single-channel statistic.*

1.4. β Pictoris: an astrophysical positive control, and the obstacle it identifies

A search that finds nothing must show that it could have found something, and the survey's strongest features show exactly that. The frozen pipeline recovers the β Pictoris circumstellar carbon monoxide blind, without being told the disc is there. It recovers it in two rotational transitions and in 9 independent execution blocks spanning 8.4 yr: 5 Band 3 windows at $T_\star$ up to 30.76, and 4 Band 6 windows at 11.68. The identification is kinematic and does not rest on those statistics. Carried through the frame chain — to the barycentre at each block's own epoch, then to the star's rest frame — all 10 crossings lie within 0.5 km s^{-1} of the stellar systemic velocity, in the two lowest CO transitions of a star with mapped circumstellar CO (??). The Band 3 peaks agree across blocks to 1.00 channel once each block's own tuning is removed, so any apparent drift is far below the grid searched. The alternative reading requires a transmitter placed by coincidence at matched offsets from two CO transitions of one star.

β Pictoris is also the control that shows the recurrence test of §1.6 working in the positive direction. Every β Pictoris carbon-monoxide feature the search flagged is recovered in an independent execution block, and the screened Band 6 window returns $T_\star = 8.95$ in the second block of its own recurrence set. A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat where one exists.

One of those blocks is a counterexample, and it is the more useful half of the control. In it the control ring is brighter than the star, and several of the 512 control positions exceed the stellar statistic, so the spatial screen rejects a window containing carbon monoxide that is known to be real and known to be centred on the star. The disc is resolved on the scale of the primary beam in that configuration, which is exactly the geometry the screen is built to reject. **The screen is therefore not a test of authenticity but a test of compactness**, and on a real, spatially extended source it fails in the conservative direction. That is a stronger statement than a clean positive control would be, because it bounds the screen's reach as well as demonstrating it (Fig. 3).

HD 48370 is the complementary case and is equally instructive. Its crossing reaches $T_\star = 26.83$, but the control ring rises with the star, to 18.36, as emission filling the primary beam does. The velocity matches the

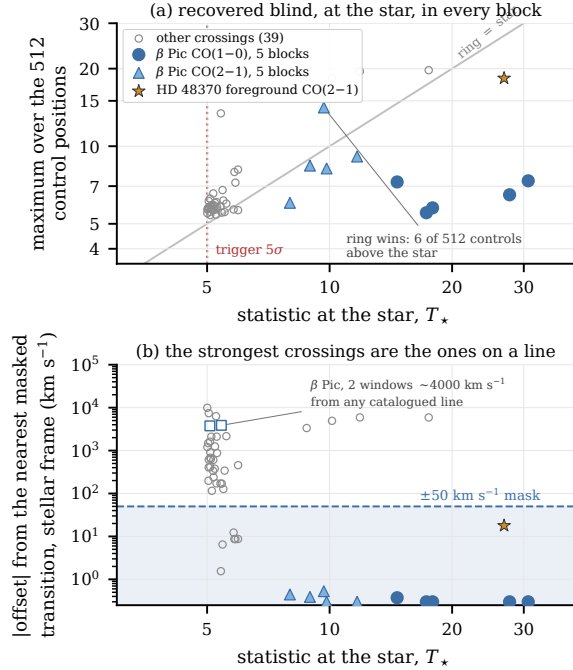


FIG. 3.— The β Pictoris positive control, in both directions. (a) The spatial screen. One point per execution block in which the blind search recovers circumstellar carbon monoxide, plotted as the stellar statistic against the maximum over that window’s 512 control positions: the 9 blocks in which the star outranks every control, and one further block in which the ring wins. *The control is that the search recovers a real narrow line at a stellar position, in two transitions and in every one of these blocks.* It is not that the star outranks the ring everywhere. In block X7116f1.X1785, Band 6, the control ring reaches 14.13 against 9.68 at the star, with 6 of the 512 control positions above it, so the screen rejects a window holding carbon monoxide that is real, centred on the star and independently mapped. The disc is resolved on the scale of the primary beam in that configuration, which is the geometry the screen is built to reject, so **the screen measures compactness and not authenticity** (§1.4). (b) Line confusion, which is what makes the screen’s job hard. Each crossing is plotted by its velocity offset from the nearest masked transition *in its own star’s rest frame*, with the adopted $\pm 50 \text{ km s}^{-1}$ half-width marked. The 10 β Pictoris carbon-monoxide crossings land $0.1\text{--}0.5 \text{ km s}^{-1}$ from CO, at the resolution of the systemic velocity itself, while the two CS(5 $\rightarrow$ 4) crossings of the same star sit $3765\text{--}3870 \text{ km s}^{-1}$ from anything catalogued and are not attributed at any width. 39 further crossings are drawn in grey, with HD 48370 marked; 52 of the 56 are plotted, the remainder being crossings for which the release stores no control ring. *A crossing is identified by where it falls in this plane and not by how strong it is*, which is why the line mask and not the trigger is the productive test.

prominent CO(2 $\rightarrow$ 1) that ? report toward the source and flag as possible foreground cloud emission. The block’s $^{13}\text{CO}(2\rightarrow 1)$ window is bright at the ring and faint at the star. Extended line emission is therefore rejected by the same spatial statistic that promotes β Pictoris.

Taken together, these two results identify the dominant obstacle for millimetre technosignature searches, and it is not sensitivity. The strongest apparent signals in this survey are circumstellar and foreground molecular emission, and 19 of the 56 crossings land on a catalogued transition. The mask that removes them costs per cent of the unique-frequency union over species, and it is nevertheless the single most productive test in the chain. The warning for future work is about line lists rather than about depth, and Appendix ?? puts a number on it.

1.5. The crossings that survive the line mask

1 of the 56 crossings is both unattributed and outranks every one of the 512 control positions in its own window: 61 Vir at 346.313811 GHz , in Band 7. The event closest to it in the ledger, CP–72 2713 at 344.269746 GHz , lies -12 km s^{-1} from SO(8 $_8$ –7 $_7$) in the star’s own frame, and the mask therefore attributes it. These are the two most interesting events in the experiment, and everything measured about them is in Table 3 rather than scattered through the text.

Both are absent when the same frequency is observed again. CP–72 2713 is the cleaner case. The next observation of that frequency began 2.05 hours later and is deeper; a carrier persisting at the discovery flux would have returned $T_* = 6.30$ there, and the value measured at the predicted cell is 3.60. That excludes a persistent source at 7.1σ — from a repeat two hours later, with no appeal to a line list, a rank or a model. 61 Vir has 3 covering repeat blocks of the 4 the catalogue records, the nearest 1.97 hours away, expected $T_* = 9.70$ against 2.71 measured: 9.3σ . Neither is carried forward.

1.5.1. The visibility fit, and why it is not the discriminant

51 of the 56 crossings were re-measured in the visibility domain, by phase rotating to the star and following the recovered drift (Table 4). The result sits beside the chain rather than inside it because a noise maximum at the stellar position has flux at the stellar position: at the trigger the fit returns a real part close to the trigger itself and adds little to the trigger that selected it. Injected sources in the bin the unattributed crossings occupy return $\text{Re}/\sigma = 6.58 \pm 1.97$, and the two events above are consistent with a real source and do not exclude one. **The discriminating evidence is the line mask and recurrence.**

1.6. Recurrence, applied to every unattributed crossing

A persistent transmitter should be there the next time the same frequency is observed. That is the confirmation criterion, applied in the star’s own frame: the expected channel follows from the Earth’s motion at the repeat epoch and the star’s systemic velocity, the discovery drift is transported to it, and the survey’s own statistic is read at that one cell. Nothing is maximised, so what comes back is an amplitude with an uncertainty.

The test is run on *every* unattributed crossing with any covering repeat coverage, and not only on those that outrank their controls: the control rank has no calibrated false-alarm probability, so it cannot be allowed to decide which crossings are tested. 36 of the 37 were tested, over 439 covering repeat windows in 146 execution blocks, 38 of them from the archival extension searched with the frozen pipeline after the statistic and the mask were fixed. *Nothing recurs.* The largest statistic found at any predicted cell is $T_* = 4.71$, and 2.48 at the matched drift, against a trigger of 5. Every tested crossing excludes a carrier persisting at its discovery flux by at least 5.0σ , median 11.6σ ; the four recurrence columns of Table 5 give the block count, the largest statistic found there at any drift, the statistic a persistent carrier would have produced and the exclusion, crossing by crossing.

One crossing could not be tested at all, and is named rather than counted: HD 14055 at 331.1559 GHz in

TABLE 3

THE TWO MOST INTERESTING EVENTS. EVERYTHING MEASURED ABOUT THE ONE UNATTRIBUTED CROSSING THAT OUTRANKS ALL 512 CONTROL POSITIONS, AND ABOUT THE ONE THE MASK ATTRIBUTES TO A SULPHUR MONOXIDE TRANSITION. T_* IF PERSISTENT IS WHAT A CARRIER AT THE DISCOVERY FLUX WOULD HAVE RETURNED AT THE PREDICTED STELLAR-FRAME CELL OF THE REPEAT COVERAGE COMBINED; T_* MEASURED IS THE LARGEST VALUE FOUND THERE; THE EXCLUSION IS THE DIFFERENCE IN UNITS OF THE COMBINED NOISE.

	61 Vir	CP-72 2713
ALMA band	7	7
Frequency (GHz)	346.313811	344.269746
Channel width (kHz)	488.28	488.28
Drift rate (Hz s^{-1})	+3182	-2056
T_*	5.958	5.809
Highest control T	5.65	5.68
Flux density (mJy)	118.8 ± 19.9	11.3 ± 2.0
EIRP (W)	5.1×10^{14}	8.9×10^{14}
Nearest transition	CO(3-2)	SO(8 ₈ -7 ₇)
Offset (km s^{-1})	+457	-12
Repeat blocks	3	5
Nearest repeat (h)	1.97	2.05
Longest baseline (d)	3	2
Covering blocks in the catalogue	4	1
T_* if persistent	9.70	6.30
T_* measured	2.71	3.60
Exclusion	9.3σ	7.1σ
Disposition	not confirmed	attributed

block X1001fdc_X29ae, whose discovery dynamic spectrum was not retained, so there is no amplitude to transport into the 30 later epochs that do cover it. A further 2 blocks the catalogue lists as covering a crossing frequency have also lost their spectra and are not counted

as repeat coverage. Nor is the archive exhausted: 18 further in-scope blocks carry no pipeline calibration product in the ALMA delivery at all (§??).

η Crv shows why the frame correction is not a detail. Its crossing at 357.281464 GHz has exactly 2 covering epochs in the archive, the second 2.19 times deeper; matching in the stellar frame moves the expected cell by -80.2 channels, so a sky-frequency test would have read the wrong channel. There the statistic is -0.618 where a persisting carrier would have given 10.95: *excluded at 11.57σ from one epoch*, carrying 4.79 equivalent discovery epochs of sensitivity.

1.6.1. What can be searched, and what can be confirmed

A sensitivity limit and a confirmation limit are different statements. Call a transmitter persistent if it would be detectable above the quoted power in every independent observation whose coverage contains its frequency. The experiment then has two extents. What it *searched* is 60 systems, 402 Class A windows and 181.5 GHz of fine-channel coverage. What it could have *confirmed* is smaller: 30 of those systems, and 79.7 GHz or 44 per cent of the Class A coverage, were observed twice at the same frequency more than a day apart; beyond a year, 11 systems and 17.8 GHz. Across the crossings tested the usable baselines run from 0.46 hours to 1.8 yr, and for 20 of 36 the nearest repeat falls within a day. **These tests exclude a carrier persistent over hours to days where there is repeat coverage; they bound no duty cycle measured in months or years, and over half the searched systems they bound nothing at all.**

TABLE 4

THE VISIBILITY-DOMAIN RE-MEASUREMENT AND THE THREE SETS OF CONTROL POSITIONS THE EXPERIMENT USES: THE RANK SCREEN COMPARES THE STAR WITH 512 POSITIONS IN THE IMAGE PLANE, FULL DYNAMIC SPECTRA ARE RETAINED FOR 8 OF THEM, AND 12 POSITIONS ARE REACHABLE IN THE VISIBILITIES.

Crossings re-measured in the visibility domain	51 of 56
consistent with a point source at the stellar position	41
of those, unattributed	31
Control positions in the image-plane rank screen	512
with a full dynamic spectrum retained	8
Control positions available in the visibility fit	12

TABLE 5

THE CROSSING LEDGER. EVERY ONE OF THE 56 THRESHOLD CROSSINGS: THE STAR, ITS EXECUTION BLOCK, THE ALMA BAND, THE CROSSING FREQUENCY AS THE CORRELATOR DELIVERED IT, THE SAME FREQUENCY IN THE STAR'S REST FRAME, THE SEARCH STATISTIC T_* , WHETHER THE STAR OUTRANKS ALL 512 CONTROL POSITIONS, THE NEAREST MASKED TRANSITION, THE STELLAR-FRAME OFFSET Δv_* FROM IT, AND THE DISPOSITION. A CROSSING IS ATTRIBUTED WHERE $|\Delta v_*| \leq 50 \text{ km s}^{-1}$; THE TRANSITION NAMED IS THE NEAREST IN THAT FRAME, SO FAR FROM EVERY TRANSITION IT IS THE MAGNITUDE AND NOT THE NAME THAT CARRIES INFORMATION. ν_* FOLLOWS FROM ν_{topo} BY THE BARYCENTRIC TERM OF THAT BLOCK'S EPOCH AND THE STAR'S SYSTEMIC VELOCITY AND NOTHING ELSE, SO THE LAST COLUMN IS REPRODUCIBLE FROM THE TWO BEFORE IT. BLOCK IDENTIFIERS OMIT THE CONSTANT A002_ PREFIX.

Star	block	B	ν_{topo} (GHz)	ν_* (GHz)	T_*	scr line	Δv_* disposition
β Pic	Xf5d76d_Xeb8	3	115.2603	115.2711	30.765	Y CO(1-0)	-0.2 attributed
β Pic	Xf5d76d_Xe19	3	115.2603	115.2711	27.684	Y CO(1-0)	-0.2 attributed
HD 48370	Xc26103_X155a	6	230.5117	230.5243	26.834	Y CO(2-1)	-17.8 attributed
β Pic	Xf4d6f_Xc7	3	115.2620	115.2711	17.907	Y CO(1-0)	-0.2 attributed
BD+05 1668	Xc7fa6f_X28a8	6	240.1016	240.1054	17.519	n CS(5-4)	-5911.9 unattributed; below rank screen
β Pic	Xf5d76d_Xcc5	3	115.2604	115.2711	17.288	Y CO(1-0)	-0.2 attributed
β Pic	Xf5d76d_X32f1	3	115.2606	115.2713	14.654	Y CO(1-0)	+0.4 attributed
BD+05 1668	Xc7fa6f_X28a8	6	224.7422	224.7458	11.868	n ¹³ CO(2-1)	+5913.1 unattributed; below rank screen
β Pic	Xd9668b_X3a90	6	230.5161	230.5378	11.681	Y CO(2-1)	-0.3 attributed
BD+05 1668	Xc7fa6f_X28a8	6	226.7422	226.7458	10.136	n CO(2-1)	-4931.4 unattributed; below rank screen
β Pic	X6f1341_X1484	6	230.5284	230.5379	9.832	Y CO(2-1)	-0.1 attributed
β Pic	X7116f1_X1785	6	230.5272	230.5384	9.677	n CO(2-1)	+0.5 attributed
β Pic	Xd9668b_Xa9df	6	230.5160	230.5377	8.948	Y CO(2-1)	-0.4 attributed
BD+05 1668	Xc7fa6f_X28a8	6	242.1953	242.1992	8.784	n CS(5-4)	-3349.2 unattributed; below rank screen
β Pic	Xa7a216_X23e4	6	230.5273	230.5377	7.988	Y CO(2-1)	-0.4 attributed
61 Vir	Xc079b5_X82f	7	346.3138	346.3232	5.958	Y CO(3-2)	+457.1 unattributed; rank-flagged
HD 285968	Xd98580_X3bf6	6	230.5021	230.5447	5.956	n CO(2-1)	+8.7 attributed
HD 285968	Xdb7ab7_X5b4e	6	230.5116	230.5447	5.868	n CO(2-1)	+8.7 attributed
HD 285968	Xdb4b9a_X108f	6	230.5100	230.5447	5.835	n CO(2-1)	+8.7 attributed
CP-72 2713	Xff0235_X4a6d	7	344.2697	344.2965	5.809	Y SO(8-7 ₇)	-12.3 attributed
HD 139084 B	Xcd8029_Xb6b0	7	345.8241	345.8241	5.763	n CO(3-2)	+24.4 attributed
HD 14055	Xff3d8_Xd46	7	331.7699	331.7664	5.575	n SO(1 ₂ -0 ₁)	+2167.0 unattributed; below rank screen
HD 23484	X10b22d7_X2445	6	230.7904	230.8006	5.520	n CO(2-1)	+341.5 unattributed; below rank screen
HD 161868	Xff45a6_X10ca8	7	345.6338	345.6491	5.481	n CO(3-2)	-127.4 unattributed; below rank screen
HIP 10679	Xa95c04_X14cd	6	220.4185	220.4035	5.456	n ¹³ CO(2-1)	+6.5 attributed
HD 14055	X11adad7_X25c5e	7	345.4564	345.4303	5.448	n CO(3-2)	-317.0 unattributed; below rank screen
HD 207129	Xc19d6f_X5706	6	230.4226	230.4070	5.433	n CO(2-1)	-170.3 unattributed; below rank screen
β Pic	Xd9668b_X3a90	6	241.7512	241.7739	5.420	n CS(5-4)	-3869.7 unattributed; below rank screen
HR 1010	Xc6fa08_X1053	6	230.5133	230.5287	5.410	n CO(2-1)	-12.0 attributed
ALMA J153702653-33192492	Xff0235_X8392	6	230.5222	230.5392	5.405	n CO(2-1)	+1.6 attributed
HD 110411	Xe45e29_Xb7d4	6	219.4538	219.4357	5.393	n C ¹⁸ O(2-1)	-170.2 unattributed; below rank screen
HD 31392	X129754b_X6946	6	231.1998	231.2231	5.289	n H30 α	-876.3 unattributed; below rank screen
HD 14055	Xff99e1_X24be0	7	331.7078	331.7035	5.277	n SO(1 ₂ -0 ₁)	+2109.7 unattributed; below rank screen
AU Mic	Xa42f75_X319	6	230.6864	230.6705	5.274	n CO(2-1)	+172.3 unattributed; below rank screen
HD 69830	X12209f7_X16edb7	7	322.4325	322.4444	5.259	n SO(1 ₂ -0 ₁)	-6317.5 unattributed; below rank screen
ALMA J153702653-33192492	Xfe62c1_X398	6	216.1844	216.2036	5.228	n SiO(5-4)	-1244.7 unattributed; below rank screen
AU Mic	X899169_X1838	6	230.8252	230.8294	5.221	n CO(2-1)	+379.0 unattributed; below rank screen
HD 14055	Xff99e1_X2c3bb	7	345.5256	345.5216	5.186	n CO(3-2)	-237.9 unattributed; below rank screen
HD 14055	Xff99e1_X3835	7	345.0099	345.0070	5.177	n SO(8 ₈ -7 ₇)	+606.4 unattributed; below rank screen
HD 14055	Xff99e1_X2b2ee	7	344.7007	344.6965	5.175	n SO(8 ₈ -7 ₇)	+336.0 unattributed; below rank screen
LHS 1140	Xd9d7c7_X6fd2	6	230.6355	230.6268	5.136	n CO(2-1)	+115.5 unattributed; below rank screen
HD 10647	Xff294f_Xa7f0	7	331.6570	331.6943	5.120	n SO(1 ₂ -0 ₁)	+2101.4 unattributed; below rank screen
CD-57 1054	Xcf525f_X5aa4	7	346.5099	346.5265	5.108	n CO(3-2)	+633.4 unattributed; below rank screen
ALMA J153702653-33192492	Xf8f6a9_X10359	6	217.7292	217.7269	5.091	n H ₂ CO(3-2)	-680.5 unattributed; below rank screen
GJ 849	Xd9668b_X841a	6	230.8769	230.8581	5.090	n CO(2-1)	+416.3 unattributed; below rank screen
ALMA J153702653-33192492	Xfe62c1_X398	6	218.8729	218.8923	5.090	n C ¹⁸ O(2-1)	-912.2 unattributed; below rank screen
HD 14055	X1001fdc_X29ae	7	331.1559	331.1546	5.082	n SO(1 ₂ -0 ₁)	+1610.2 unattributed; below rank screen
β Pic	Xd9668b_Xa9df	6	241.8367	241.8594	5.079	n CS(5-4)	-3765.1 unattributed; below rank screen
ALMA J153702653-33192492	Xf8f6a9_X10359	6	218.6711	218.6688	5.048	n H ₂ CO(3-2)	+613.5 unattributed; below rank screen
HD 69830	X12277ed_X178c6	7	321.2005	321.2149	5.046	n SO(1 ₂ -0 ₁)	-7436.5 unattributed; below rank screen
HR 1010	Xc66ce1_Xaa9	6	230.8348	230.8491	5.044	n CO(2-1)	+404.6 unattributed; below rank screen
61 Vir	Xc067f7_X822	7	345.5599	345.5678	5.038	n CO(3-2)	-197.8 unattributed; below rank screen
HD 170773	Xff0235_X8c42	7	331.0101	331.0234	5.029	n SO(1 ₂ -0 ₁)	+1490.8 unattributed; below rank screen
ALMA J153702653-33192492	Xd9d7c7_Xbb64	7	347.2179	347.1890	5.009	n CO(3-2)	+1207.7 unattributed; below rank screen
HD 14055	X1190de8_Xd526	7	344.4676	344.4515	5.006	n SO(8 ₈ -7 ₇)	+122.7 unattributed; below rank screen
η Crv	X122b6ff_X1041e	7	357.2815	357.2475	5.006	n CO(3-2)	+9928.0 unattributed; below rank screen

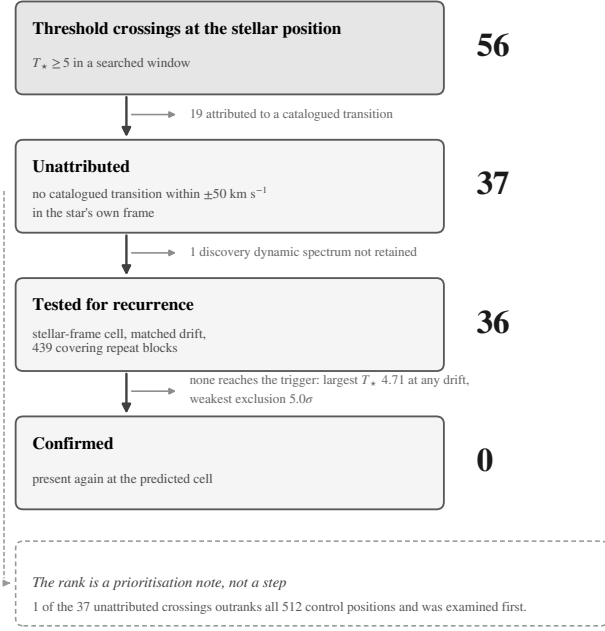


FIG. 4.— What disposes of a threshold crossing. 56 crossings; 19 attributed to a catalogued transition by the stellar-frame mask; 37 unattributed, of which 36 are tested for recurrence against 439 covering repeat windows and none recurs. The control rank is an annotation and not a step: it says which crossing was examined first, not which survives. Every count is read from the ledger, so the figure and Table 5 cannot disagree.